One responsibility.
One state owner.
Many possible backends.

Omega Hive × ASI:chain

From a module map to a working storage, messaging and execution slice

M0 + M1 COMPLETE **M2 NOT STARTED**

Architecture review • 16 October 2026 • proposed v0 contracts

The architecture separates three commitments

A function boundary, an authority boundary, and an evidence boundary.

01

Replace behavior safely

Select a versioned contract and semantic profile. Keep the consumer independent of the provider.

02

Write through the owner

Episodes, beliefs, attention and jobs each have a single logical authority for their state.

03

Name what was observed

Storage, delivery, execution and finality carry separate receipts and explicit limits.

M1 demonstrates one narrow path through that architecture

STORE → SHARE REF → FETCH → DELEGATE → RECOVER → COMMIT DIGEST

Fifteen modules; four responsibility groups

Logical boundaries are independent of process and repository boundaries.

CLIENT / COGNITION

- Episodic memory
- Belief revision
- PLN kernel
- ECAN / attention
- Inference control

SHARED APPLICATION PORTS

- AtomStorage
- Relations · spec
- Search projection · spec
- Agent messaging
- Delegated execution

CONTROL & POLICY

- Registry
- Compatibility resolver
- Authorization

Local harness policy and static descriptors

PROVIDER ADAPTERS

DA storage

Real DeTTa libraries

Consensus / finality

Real isolated F1R3 node

SPECIFICATION ONLY

IMPLEMENTED HARNESS

REAL PROVIDER INTEGRATION

Data responsibilities: bytes → references → history

The first five modules separate persistence, identity, structure, discovery and provenance.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / STATUS
DA storage	Persist opaque bytes put · get · availability	Payloads and shares REAL LOCAL PROVIDER
AtomStorage	Assign scoped immutable references create · get · getMany	Reference + replay metadata IMPLEMENTED
Relations	Represent ordered structural roles assertRelation · getRelation	Immutable relation records SPECIFIED
Search projection	Find references at a known watermark query	Rebuildable index SPECIFIED
Episodic memory	Preserve ordered events + provenance recordEpisode · appendEvent	Private episode history SPECIFIED · M2

Capture and retrieve data before choosing its meaning.

Cognitive responsibilities: history ≠ truth ≠ attention

All four modules below are typed specifications. No cognitive provider runs in M1.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / EXCLUSION
Belief revision	Revise under an explicit numerical model submitEvidence · readBelief	Belief + evidence ledger No agenda or attention writes
PLN kernel	Apply a rule to explicit premises profiles · applyRule	No mutable domain state Returns a proposed derivation
ECAN / attention	Allocate, select and decay attention submitStimulus · select · advance	Attention balances + logical time No PLN dependency
Inference control	Schedule bounded reasoning work step	Step receipts + checkpoint No ownership of truth or attention

Mathematics and attention policy remain explicit review gates.

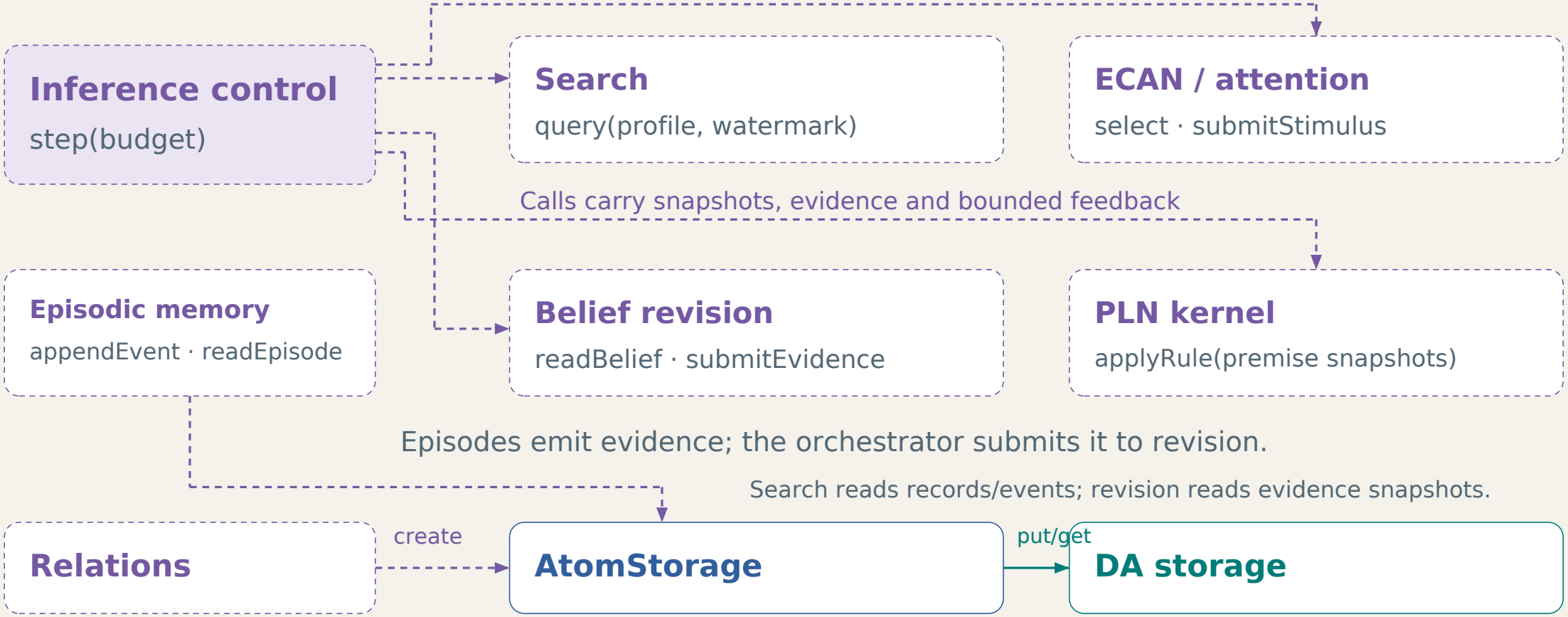
Control responsibilities: move work without moving authority

The remaining six modules connect owners and report exactly what happened.

MODULE	FUNCTION / PRIMARY OPERATIONS	OWNED STATE / STATUS
Messaging	Accept, deliver, acknowledge send · poll · ack	Inbox + deduplication · IMPLEMENTED
Execution	Run one named operation submit · status · result	Jobs + completion outbox · IMPLEMENTED
Registry	Describe configured providers descriptors	Static configuration · IMPLEMENTED
Resolver	Match required semantics resolve	No mutable state · IMPLEMENTED
Consensus / finality	Submit approved commitment; observe submit · observe	Signed intent journal · REAL LOCAL CHAIN
Authorization	Check current caller + scope + method authorize / live policy checks	Server-owned policy · LOCAL FIXTURE

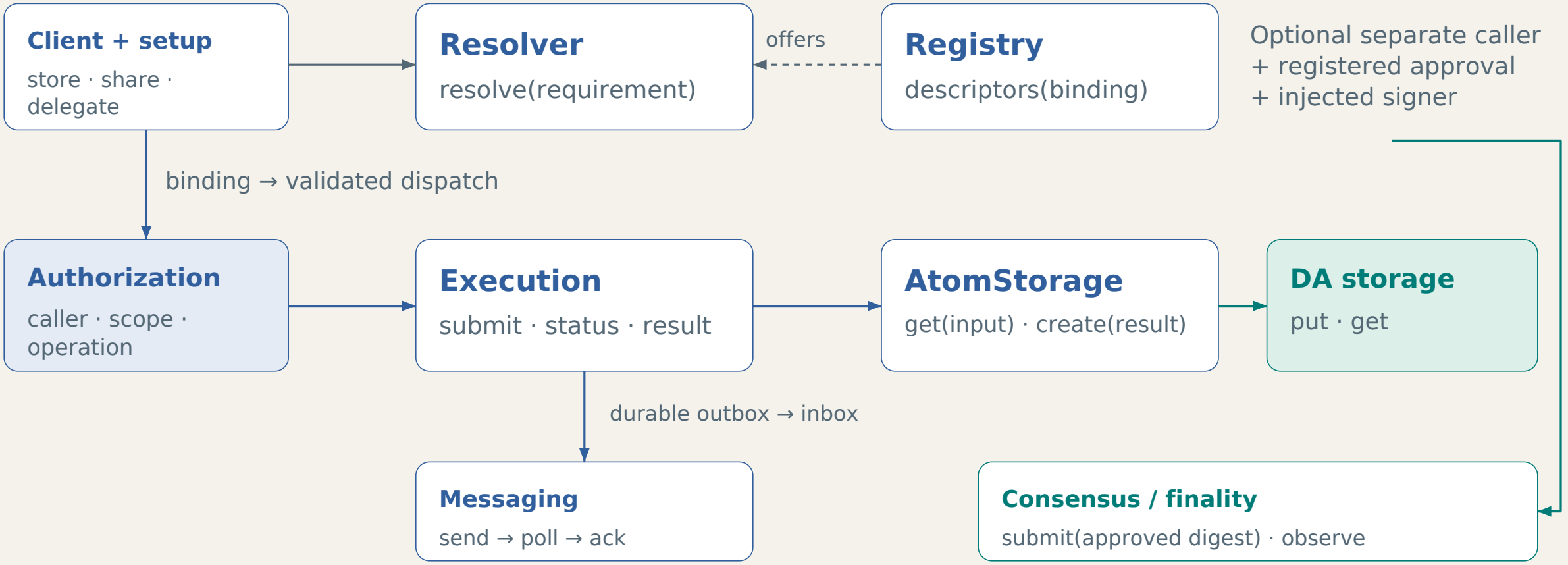
Function dependencies: the proposed cognitive path

Dashed arrows = specified calls. Solid arrow = implemented call. Direction = caller → provider.



Function dependencies: admission, work and delivery

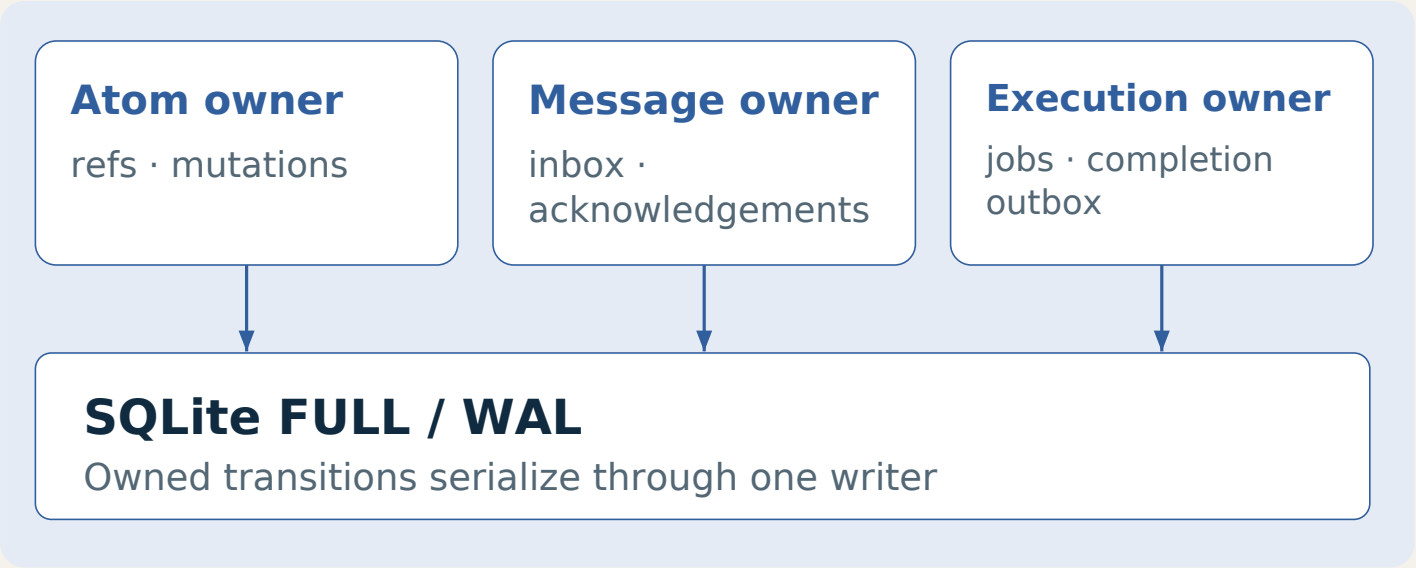
Resolve once per configured capability; authorize every invocation.



State ownership survives co-location

A shared database is a deployment choice. Its tables still have different writers.

M1: TRUSTED LOCAL SERVICE



Payload authority

DeTTa FileStorage owns the bytes and erasure shares.

Future private client state

Episodes, beliefs and attention stay behind their respective owner ports.

External callers receive methods and receipts; they receive no table or owner setters.

Keep four architecture choices independent

Changing placement must not silently change the meaning of an operation.

INTERFACE	**What behavior is promised?**	atom.storage / immutable-bytes/1
IMPLEMENTATION	**Who provides that behavior?**	AtomOwner + injected DeTTa adapter
TRANSPORT	**How does a caller cross the boundary?**	In-process call or private JSON stdio
PLACEMENT	**Where does the implementation run?**	Trusted local service; optional chain adapter

Only composition imports concrete adapters. Domain contracts stay backend-neutral.

Compatibility is a semantic filter

A matching name is only the start of provider selection.

Version

Explicit same-major range

→

Profile

Exact semantics + features

→

Operations

Required calls available

→

Guarantees

Durability, privacy, limits

Required proof? Fail closed.

No verifier is installed. A proof-shaped field cannot satisfy a verification requirement.

Actual wire version: 1.0.0

Additive 1.2.3 compatibility is a resolver fixture, not an available wire provider.

Identity, replay and authority travel separately

The invocation names an operation; trusted context identifies the caller.

```
interface_ref: atom.storage@1.0.0
profile: immutable-bytes/1
operation: create
scope_ref: shared-test
message_id: delivery-042
idempotency_key: capture-007
body: {content_type, bytes, sharing}
```

Illustrative field view · not a literal wire request

Occurrence ≠ content

Identical bytes can be distinct observations.

Retry ≠ new mutation

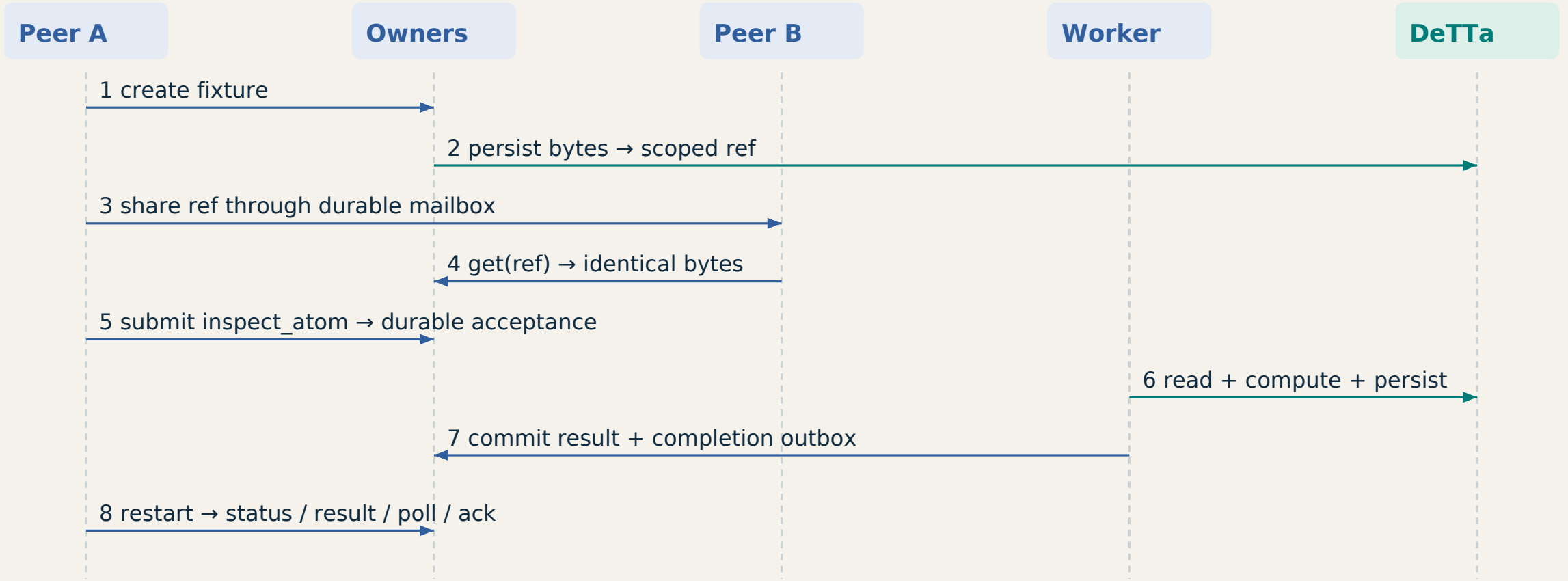
Same key + semantic input recovers the same effect.

Scope identifier ≠ permission

Live policy uses the authenticated principal.
Correlation + reply route make async work traceable.

One slice crosses real process and storage boundaries

Two independent protocol peers. One allowlisted operation. Two interchangeable service bindings.



A receipt proves one fact at one boundary

These observations are independent. They are not one universal success state.

STORAGE

Payload persisted

Local shares; declared durability

MESSAGING

Message accepted

Delivery repeats until recipient ack

EXECUTION

Result committed

Job result + completion outbox

CHAIN SUBMISSION

Signed intent accepted

Deploy ID; outcome still pending

CHAIN INCLUSION

Deploy observed in block

Block ID; effects may be pending

CHAIN FINALITY

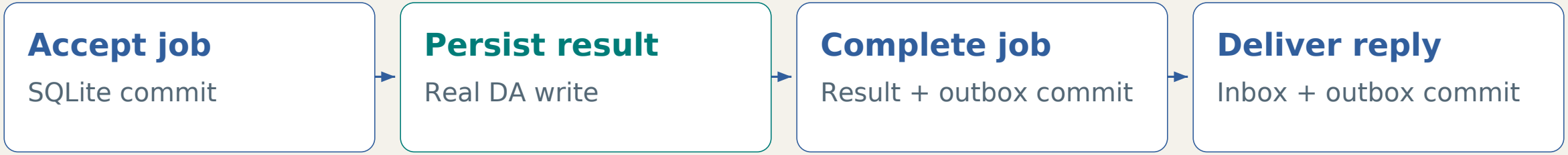
Effects reported finalized

Node observation + evidence

Finality does not establish indefinite availability of off-chain bytes.

Crash recovery is a replay protocol

SQLite and DA do not share an atomic commit. The gap is handled explicitly.



SIGKILL A

Before acceptance reply

Retry recovers the accepted job.

SIGKILL B

After DA write

Replay reuses the same coordinate; an unreferenced artifact may remain.

SIGKILL C

Before dispatch

The persisted outbox survives and later reaches the inbox.

One logical result; physical computation may repeat. Tested failure model: same-host process death.

The DA boundary carries bytes, not cognition

The application adapter is outside the unchanged DeTTa core.

Application contracts

AtomRef · scope · semantic replay

→

External adapter

Opaque payload + generic coordinate

→

DeTTa core

DA encoding + FileStorage

4 data shares + 2 parity shares

S1	S2	S3	S4	S5	S6
present	present	present	present	removed	removed

Observed reconstruction

Cached payload + 2 shares removed: exact bytes recovered. Third missing share: unavailable.

No Omega IDs, truth values, ECAN balances or job state inside the generic DA schema.

Authorization is checked where state is owned

Authenticated context → current policy → explicit method → owner transition.

ATTEMPT	OBSERVED BEHAVIOR	BOUNDARY
Read another scope	UNAUTHORIZED	Scope IDs confer no authority
Poll or ack another route	UNAUTHORIZED	Recipient owns delivery state
Reuse key with different input	CONFLICT	Semantic idempotency
Call after policy revocation	Rejected on next checked call	Policy is live, including retries
Overfill job queue / inbox	OVERLOADED / retained outbox	No silent acknowledged-work drops

Local fixture policy; trusted OS user and files. Production SCS enforcement remains open.

What ran, what was a fixture, what is unavailable

Seven requested repositories inspected; only demonstrated integrations are marked real.

COMPONENT	CLASSIFICATION	EVIDENCE / LIMIT
DeTTa DA + FileStorage	REAL PROVIDER LIBRARIES	Persisted shares + actual reconstruction
F1R3node / ASI seam	REAL ISOLATED CHAIN	Signed deploy; inclusion + effect finality
Peer A / Peer B / inspect_atom	PROTOCOL + OPERATION FIXTURES	Real separate worker; no reasoning
Policy + chain unit responses	EXPLICIT FIXTURES	Revocation / failure mapping / timeout tests
Omega Hive	RUNTIME UNAVAILABLE	Onboarding index present; Hive checkout absent
DAS / MeTTaClaw / ECAN / PLN	INSPECTED; NOT INTEGRATED	Candidate seams and future adapters recorded

A real result digest reached a real local chain

One registered approved commitment. Durable signed intent. Direct node observations.

PENDING
Signed deploy accepted

→

INCLUDED
Block observed; effects pending

→

FINALIZED
Deploy effects reported finalized

Recorded inclusion

Block height	**3**
Execution error	**false**
Rejection count	**0**
Execution cost	**351 phlo**

Commitment = persisted result digest

```
f9236451ff785dc58315251cf864a3d1
0938412551cf45f448055061bc953ea1
```

Observation window: 4.013 s
One local sample; no latency SLA

Only the generated result digest entered Rholang. Full payload stayed in local DA.

Evidence concentrates on boundaries that can fail

Recorded passing tests and experiments; future cognitive test names are not included.

27

contract + adapter tests

102

upstream DA / storage tests

2

bindings; same consumer

Identity + isolation

Exact bytes, immutable revision, scoped routes, live revocation.

Admission + compatibility

Bounded overload, deadlines, versions, profiles and required features.

Replay + recovery

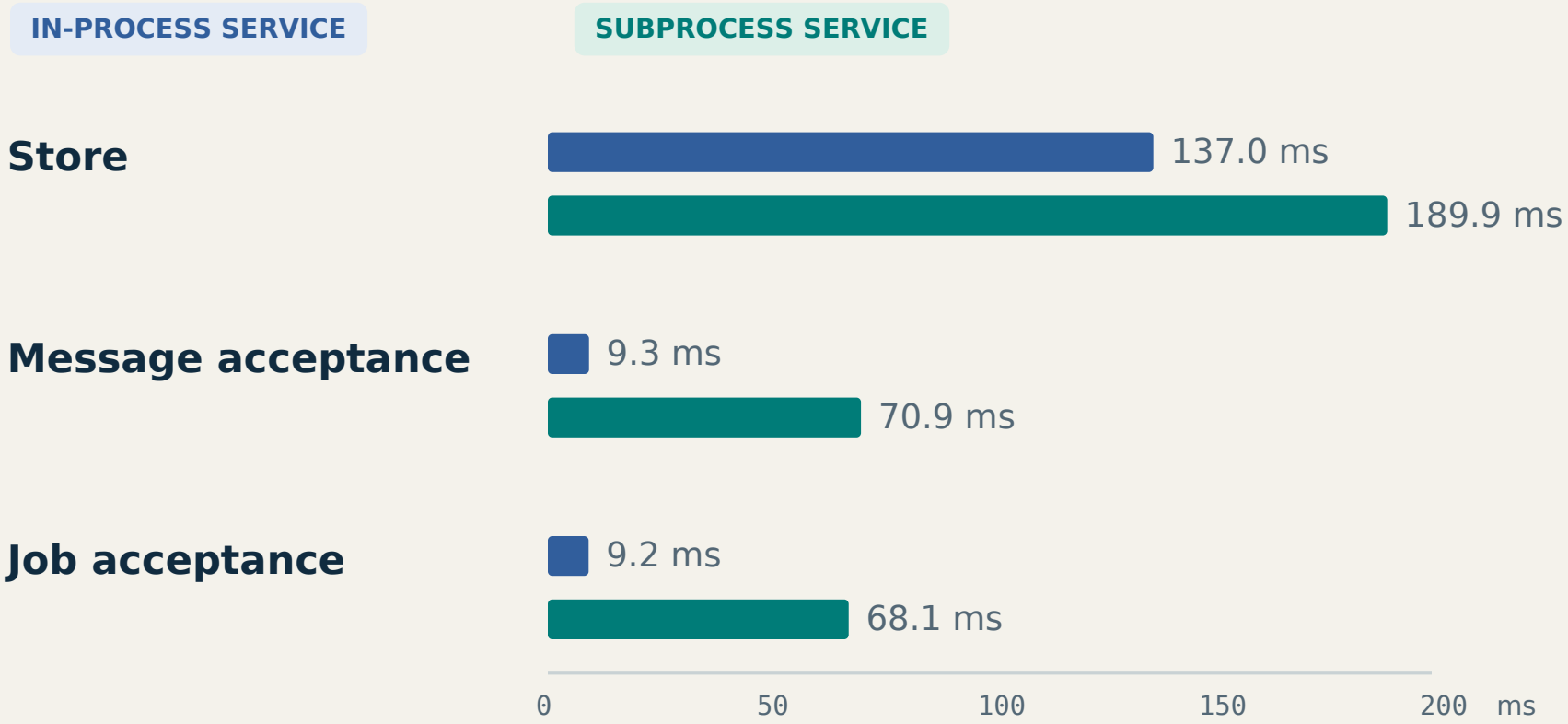
Conflicting keys, concurrent retries, three SIGKILL boundaries.

Provider behavior

Real erasure recovery, corrupt shares, chain mapping and signed-intent retry.

A small baseline, with the failure path included

One 62-byte fixture, one slice per binding. Bars share a zero baseline; this is not a benchmark.



Entire slice

0.756 s / 1.761 s
Includes two worker kills and recovery.

0 observed loss

Acknowledged data; tested same-host SIGKILL model only.

Both bindings still use real DA subprocesses and separate workers. No scalability conclusion.

The design trades breadth for inspectable guarantees

Accepted prototype choices; community models and production guarantees remain unresolved.

DECISION	WHAT IT MAKES EXPLICIT	COST / OPEN BOUNDARY
Immutable scoped references	Identity and replay are separate	No GC, merge or mutable atom revisions
One writer + durable outbox	Recovery has a clear authority	Small local harness; no distributed scheduler
Exact semantic profiles	Provider swaps preserve behavior	Numerical models require specialist review
Generic DA + external adapter	Cognition stays above storage	No distributed custody or lifetime guarantee
Approved digest on chain	Finality evidence stays separate	No arbitrary deploy or private episode upload

M0 and M1 are complete. The next gate stays closed.

The architecture describes the next steps; this implementation stops before M2.

M0

Contracts

Discovery, ownership, typed ports, conformance obligations.

M1

Real local slice

Store, share, inspect, recover; optional chain commitment.

M2

Episodes + revision

Private ordering, evidence lineage, reviewed numerical model.

M3 / M4

Cognition + Hive

Bounded attention / inference; selected workflow integration.

Open gates

Integration

Intended Hive runtime / version; production authorization.

Guarantees

Distributed custody; multi-validator operation; verified proofs.

Cognition

Reviewed revision model; STI policy; specialist ownership.

Modularity is a promise about behavior and authority.

The working slice makes that promise reviewable.

01 State changes through its owner.

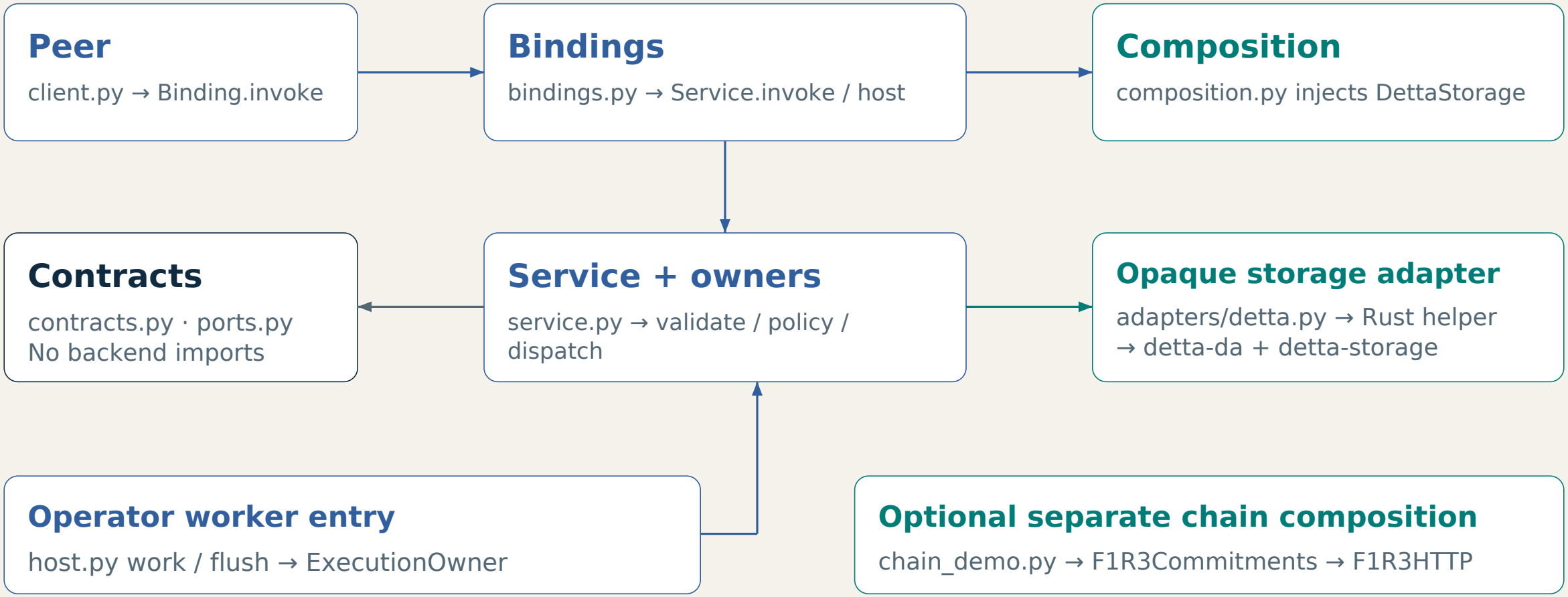
02 Providers are selected by declared semantics.

03 Receipts say precisely what was observed.

M0 + M1 delivered • No M2 implementation • Evidence and reproduction in the repository

Implementation dependency map

Solid lines identify actual calls/import composition; file names locate the corresponding seam.



Specified cognitive contracts carry test obligations

These are future acceptance cases, not executed M1 tests.

CONTRACT	SEMANTIC DECISION	NAMED CONFORMANCE OBLIGATIONS
Relations / Search	Ordered roles; dialect + watermark	REL-order / scope; SEARCH-watermark
Episodes · M2	Occurrence identity; sequence CAS	EP-order / duplicate / contrary / privacy
Belief revision · M2	Numerical model; lineage; independence	BR-overlap / cycle / contradiction / golden
Attention · M3	STI policy; logical time; budget; ties	ECAN-no-PLN / owner / budget / decay
PLN kernel · M3	Rule version; ordered premise snapshots	PLN-order / golden / purity / lineage
Inference control · M3	Candidate, rule, feedback, time limits	IC-bounds / feedback / replay / swap

Typed failures preserve the caller’s recovery choices

Unknown outcome is different from known rejection; an empty inbox is different from failure.

OBSERVATION / ERROR	MEANING	CALLER ACTION
NOT_FOUND	No such scoped atom metadata	Correct the reference
NOT_AVAILABLE	Known data / result temporarily unavailable	Retry or query the owner
OVERLOADED	Capacity or current writer lock unavailable	Back off; reuse key + semantic input
CONFLICT	Same key changed meaning / revision mismatch	Reconcile; do not blindly replay
UNAUTHORIZED / UNSUPPORTED_*	Access or contract requirements fail	Change authorization or requirements
Transport loss	Acceptance may already be durable	Recover by key / job; do not assume failure

Provenance includes working-tree state

Exact paths, remotes and source fingerprints are recorded in evidence/sources.json.

REPOSITORY	INSPECTED COMMIT	USE / SOURCE STATE
omegahive-start-here	0b15d2bed730	Clean · onboarding only; runtime absent
metta-attention	3e1c8147dc17	Clean · ECAN interfaces inspected
das	8afab5daa635	Clean · atom/query proxies inspected
mettaclaw	d2da47207fec	Existing untracked artifacts · interfaces read
reachability-atomspace-prototype	dd56f3f053ed	Clean · ownership / PLN seams inspected
secured-contract-spaces	272888f264dd	Existing modifications · linked + tested
f1r3node-rust	9ec1096e1077	Existing modifications · built + executed

Reproduce the gates; inspect the evidence

Run from the fusion-module-interfaces checkout; see runbook.md for prerequisites and exact commands.

LOCAL M0 / M1 GATE

```
uv run --python 3.11 python scripts/verify.py
```

Builds DeTTa adapter offline, runs 27 tests, executes both demo bindings.

Optional real chain gate

scripts/build_chain.py
scripts/chain_demo.py
Pinned extra tools + isolated local genesis; exact invocation in runbook.

Review artifacts

evidence/demo.json + chain-demo.json
tests.log + detta-tests.log
sources.json + workspace-files.json

Requires inspected sibling checkouts and cached Cargo dependencies. No production keys or private episodes.